Chinmoy Bhattacharjee

Junior Professor of Mathematical Stochastics · Universität Hamburg

Personal Details

Place | Date of Birth:	Kolkata, India | 6 December 1990
Nationality:	Indian
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Academic Positions

Sep. 2026–present	**Junior Professor (W1)** at Department of Mathematics, University of Hamburg
Jun. 2026	**Positive interim evaluation** (Habilitationsäquivalent) Extension of Junior Professorship until Aug. 2029
Sep. 2023–Aug. 2026	**Junior Professor (W1)** [first phase] at Department of Mathematics, University of Hamburg
Aug. 2020–Aug. 2023	**Postdoctoral Researcher** at Department of Mathematics, University of Luxembourg
Aug. 2018–Jul. 2020	**Postdoctoral Researcher** at Institut für Mathematische Statistik und Versicherungslehre (IMSV), University of Bern
Jul. 2018–Aug. 2018	**Visiting Scientist** at Indian Statistical Institute, Stat-Math Unit, Kolkata

Education

Aug. 2013–May 2018	**PhD in Applied Mathematics**, Department of Mathematics, University of Southern California Advisor: Prof. Larry Goldstein Thesis: Stein's method and its applications in strong embeddings and Dickman approximations
Aug. 2011–May 2013	**Master of Statistics (M. Stat.)**, Indian Statistical Institute, Kolkata Specialization: Mathematical Statistics and Probability
Aug. 2008–May 2011	**Bachelor of Statistics with Honours (B. Stat. (Hons.))**, Indian Statistical Institute, Kolkata

PREPRINTS

1. **C. Bhattacharjee**, A. Gusakova, and C. Hirsch. Functional central limit theorems for Laguerre tessellations. Submitted, 2026+. arXiv: 2610.06372

2. **C. Bhattacharjee**. Dickman approximation of randomly weighted sums via Stein's method. Submitted, 2026+. arXiv: 2606.27875

3. **C. Bhattacharjee** and R. O'Clarey. Chaos and superconcentration for Poisson functionals with applications in stochastic geometry. Major revision in *Electronic Journal of Probability*, 2026+. arXiv: 2603.23053

4. **C. Bhattacharjee** and A. Gusakova. Gaussian approximation for extreme points in Laguerre tessellations. Submitted, 2025+. arXiv: 2510.21665

5. **C. Bhattacharjee**, G. Peccati, and D. Yogeshwaran. Spectra of Poisson functionals and applications to continuum percolation. Submitted, 2024+. arXiv: 2407.13502

6. Z. Shi, **C. Bhattacharjee**, K. Balasubramanian, and W. Polonik. Multivariate Gaussian approximation for random forest via region-based stabilization. Submitted, 2024+. arXiv: 2403.09960

7. **C. Bhattacharjee**, A. Ilienko, B. Jin and I. Molchanov. From Gaussian to stable and back: non-Pareto points in growing dimension. Submitted, 2026+. arXiv: 2611.01096

PEER-REVIEWED PUBLICATIONS

8. **C. Bhattacharjee**, R. Versendaal, and A. Wade. Perimeter length of the convex hull of Brownian motion in the hyperbolic plane. *Journal of Applied Probability*, published online, 2026.

9. Z. Shi, **C. Bhattacharjee**, K. Balasubramanian, and W. Polonik. Gaussian and bootstrap approximation for matching-based average treatment effect estimators. *Annals of Statistics*, 54(4), 2164–2190, 2026.

10. **C. Bhattacharjee** and M. Schulte. Dickman approximation of weighted sums of independent random variables in the Kolmogorov distance. *Annals of Applied Probability*, 35(5), 3271–3309, 2025.

11. **C. Bhattacharjee**, I. Molchanov, and R. Turin. Central limit theorem for a birth–growth model with Poisson arrivals and random growth speed. *Advances in Applied Probability*, 56(3), 1004–1032, 2024.

12. **C. Bhattacharjee** and I. Molchanov. Gaussian approximation for sums of region-stabilizing scores. *Electronic Journal of Probability*, 27, paper no. 111, 2022.

13. **C. Bhattacharjee**. Gaussian approximation for rooted edges in a random minimal directed spanning tree. *Random Structures & Algorithms*, 61(3), 462–492, 2022.

14. **C. Bhattacharjee** and M. Schulte. Large degrees in scale-free inhomogeneous random graphs. *Annals of Applied Probability*, 32(1), 696–720, 2022.

15. **C. Bhattacharjee** and I. Molchanov. Convergence to scale-invariant Poisson processes and applications in Dickman approximation. *Electronic Journal of Probability*, 25, paper no. 79, 2020.

16. **C. Bhattacharjee** and L. Goldstein. Dickman approximation in simulation, summations and perpetuities. *Bernoulli*, 25(4A), 2758–2792, 2019.

17. **C. Bhattacharjee** and L. Goldstein. On strong embeddings by Stein's method. *Electronic Journal of Probability*, 21, paper no. 15, 2016.

GRANTS

2024–2027 (36 MONTHS)
Superconcentration in random geometric structures
DFG Priority Program 'Random Geometric Systems' (SPP 2265)
ROLE: Principal Investigator
AMOUNT: 219,966 Euros

2026–2028 (24 MONTHS)
Sharp noise sensitivity & exceptional times in continuum percolation
Programmes for Project-Related Personal Exchange (PPP) [DST & DAAD]
ROLE: co-PI, together with Prof. D. Yogeshwaran (ISI, Bangalore, India)
AMOUNT: 11,308 Euros

INVITED LECTURES (IN LAST FIVE YEARS)

DEC. 2026 **Workshop on Stein's Method and Its Statistical Applications**, ESSEC Business School, Cergy, France (upcoming)

AUG. 2026 **European Meeting of Statisticians (EMS)**, invited session *Stein-based inference*, Università della Svizzera italiana, Lugano, Switzerland

JUL. 2026 **Second Workshop on Stein's Method and Networks**, University of Oxford, UK

NOV. 2025 **Mathematics Colloquium**, Institut für Mathematik, Universität Osnabrück, Germany

OCT. 2025 **Workshop on Random Geometry, Graphs and Extremes**, IUC Dubrovnik, Croatia

APR. 2025 **Stochastics Seminar**, Institut for Matematik, Aarhus Universitet, Denmark

OCT. 2024 **Workshop on Random Geometry and Particle Systems**, IMT Nord Europe, Lille, France

AUG. 2024 **Bernoulli–IMS World Congress in Probability and Statistics**, invited session *Stein's method and random graphs*, Ruhr-Universität Bochum, Germany

JUN. 2024 **Workshop on Stein's Method and Networks**, University of Oxford, UK

JUN. 2024 **Workshop on Random Geometric Systems** (SPP 2265), Technische Universität Dortmund, Germany

APR. 2024 **Stochastics Seminar**, Institut für Mathematische Stochastik, Universität Münster, Germany

SEP. 2022 **Mini-Workshop: Limit Theorems for Spatial Random Structures**, Ruhr-Universität Bochum, Germany

JUN. 2022 **Stein's Method: The Golden Anniversary**, Institute for Mathematical Sciences, National University of Singapore

Teaching Experience

University of Hamburg	**Stochastische Prozesse** (current, in German) Winter 2026/27
	Stochastik 1 für Studierende der Informatik (in German) Summer 2025, Summer 2026 (jointly with Prof. Dr. Johannes Lederer)
	Seminar on Probabilistic Methods in Combinatorics Summer 2025, Summer 2026
	Stochastik 2 für Studierende der Informatik (in German) Winter 2025/26 (jointly with Prof. Dr. Johannes Lederer)
	Seminar on Percolation Theory Winter 2025/26
	Introduction to Random Graphs Master in Mathematics; Winter 2023/24, Winter 2024/25
	Introduction to Stein's Method Master in Mathematics; Summer 2024
University of Luxembourg	**Probability and Statistics II** BMATH fourth semester; Spring 2023, Spring 2022
	Refresher in Probability Master of Data Science first semester; Fall 2022, Fall 2021
	Probabilistic Methods in Combinatorics BMATH fifth semester; Fall 2021, Fall 2020
University of Bern	**Introduction to Random Graphs** MMATH second semester; Spring 2019
University of Southern California (Assistance)	**Math 125 - Calculus I**; Fall 2017 **Math 308 - Statistical Inference and Data Analysis II**; Spring 2016 **Math 307 - Statistical Inference and Data Analysis I**; Fall 2015 **Math 226 - Calculus III**; Fall 2014, Spring 2015

Supervision

PhD.	Rowan O'Clarey (Oct. 2024 – Sep. 2027 expected) -Funded by DFG grant no. 531540467 Topic: Superconcentration in random geometric structures
Other Supervision	Amélie Boussemart (Master's thesis) Oliver Gertke (Master's thesis) Luzie Grutke (Bachelor's thesis, ongoing) Maika Lu Beecken (Bachelor's thesis)

Organization & Academic Service

Workshop Organization:

MAR. 2025 **Session Co-chair:** *Spatial stochastics, disordered media, and complex networks*, in the *German Probability and Statistics Days*, Dresden, Germany.
Co-chaired with B. Jahnel (TU Braunschweig/WIAS Berlin)

SEP. 2024 **Workshop Co-organizer:** *Workshop on Stein's method and stochastic geometry*, Hamburg, Germany.
Co-organized with M. Schulte (TUHH, Hamburg)

Peer Reviewing:

JOURNALS Annals of Statistics (AoS), Applied Probability Journals, Bernoulli, Extremes, Electronic Journal of Probability (EJP), Journal of the American Statistical Association (JASA), Probability Theory & Related Fields (PTRF), Random Structures & Algorithms (RSA), Statistics & Probability Letters, Stochastic Processes & their Applications (SPA)

Thesis Examination:

2024, 2026 **Co-examiner** for two Master's theses, Universität Hamburg

2026–2027 **Co-examiner** for a doctoral thesis, Technische Universität Hamburg

Language Skills

Bengali (mother tongue), English (C2), German (B2), French (A1), Hindi (C1)